

UNIT-I: LOGIC AND PROOF TECHNIQUES

3. PREDICATE CALCULUS (FIRST-ORDER LOGIC)

3.1 Introduction to Predicate Logic

In propositional logic, statements are treated as indivisible units. For example:

- 5 is a prime number.
- It is raining.
- Ram passed the examination.

Each statement is either true or false.

However, many mathematical statements involve variables and cannot be expressed adequately using propositional logic.

Consider the following statements:

- Every student passed the examination.
- Some students own laptops.
- Every prime number greater than 2 is odd.
- There exists an integer whose square is 25.

These statements contain variables and require a more powerful logical system.

To handle such statements, we use **Predicate Logic**, also known as **First-Order Logic**.

3.2 Predicate Logic

Predicate Logic extends propositional logic by introducing:

1. Variables
2. Predicates
3. Quantifiers
4. Functions
5. Relations

Predicate logic allows us to describe properties of objects and relationships among objects.

Components of Predicate Logic

A predicate logic statement may contain:

- Variables ($x, y, z, \dots$)
 - Constants ($1, 2, 5, a, b, \dots$)
 - Predicates ($P(x), Q(x,y)$)
 - Quantifiers ($\forall, \exists$)
 - Logical Connectives ($\wedge, \vee, \neg, \rightarrow, \leftrightarrow$)
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3.3 Predicates

Definition

A predicate is a statement involving one or more variables that becomes a proposition when values are assigned to those variables.

A predicate is sometimes called a propositional function.

Example 1

$P(x): x > 5$

This is not a proposition because the value of x is unknown.

When $x = 8$,

$P(8): 8 > 5$

which is True.

When $x = 2$,

$P(2): 2 > 5$

which is False.

Example 2

$Q(x): x$ is an even number.

$$Q(4) = \text{True}$$

$$Q(9) = \text{False}$$

Example 3

$$R(x): x^2 = 25$$

$$R(5) = \text{True}$$

$$R(-5) = \text{True}$$

$$R(4) = \text{False}$$

3.4 Predicates with Two Variables

A predicate may contain more than one variable.

Example

$$P(x,y): x > y$$

Examples:

$$P(5,2) = \text{True}$$

$$P(2,5) = \text{False}$$

$$P(3,3) = \text{False}$$

Example

$$Q(x,y): x + y = 10$$

$$Q(6,4) = \text{True}$$

$$Q(3,8) = \text{False}$$

3.5 Predicates with Three Variables

Predicates may contain three or more variables.

Example

$$R(x,y,z): x + y = z$$

Examples:

$$R(3,4,7) = \text{True}$$

$$R(2,-1,5) = \text{False}$$

$$R(5,5,10) = \text{True}$$

Important Observation

$R(x,y,z)$ is not a proposition because variables remain unspecified.

However,

$$R(3,4,7)$$

is a proposition because all variables have been assigned values.

3.6 Domain of Discourse

Definition

The set from which variables take their values is called the domain of discourse or universe of discourse.

Common Domains

Natural Numbers

$$\mathbb{N} = \{1,2,3,4,\dots\}$$

Integers

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

Real Numbers

$\mathbb{R}$

Positive Integers

$$\{1, 2, 3, \dots\}$$

Students in a class

Employees in a company

Vertices of a graph

Importance of Domain

The truth value of a predicate often depends on the domain.

Example

$$P(x): x^2 = 4$$

Domain = Integers

Solutions:

$$x = 2$$

$$x = -2$$

Domain = Natural Numbers

Solution:

$$x = 2 \text{ only}$$

Domain = Real Numbers

Solutions remain:

$$x = \pm 2$$

Thus, changing the domain may change the meaning and truth of statements.

3.7 Open Sentences

An open sentence is a statement containing variables whose truth value is not yet determined.

Example

$$x + 3 = 7$$

This is an open sentence.

It becomes:

True when $x = 4$

False when $x = 2$

Example

$$x > y$$

Its truth depends on the values of x and y .

3.8 Well-Formed Formulas (WFF)

A well-formed formula is a logically meaningful expression constructed according to the rules of logic.

Examples of Well-Formed Formulas

$$P(x)$$

$P(x) \wedge Q(x)$

$P(x) \rightarrow Q(x)$

$\forall x P(x)$

$\exists x Q(x)$

Not Well-Formed

$\wedge P(x)$

$\rightarrow \rightarrow P(x)$

These violate the rules of logical syntax.

3.9 Quantifiers

Predicates become complete mathematical statements through the use of quantifiers.

Quantifiers tell us how many elements satisfy a predicate.

The two fundamental quantifiers are:

1. Universal Quantifier
 2. Existential Quantifier
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3.10 Universal Quantifier

Symbol

$\forall$

Read as:

"For all"

"For every"

Form

$$\forall x P(x)$$

Meaning:

$P(x)$ is true for every element in the domain.

Example 1

For every real number x ,

$$x^2 \geq 0$$

This statement is True.

Reason:

The square of every real number is non-negative.

Example 2

For every integer x ,

x is even.

False.

Counterexample:

$$x = 3$$

Example 3

For every positive integer n ,

$$n + 1 > n$$

True.

3.11 Counterexamples

To show that a universal statement is false, it is sufficient to find one counterexample.

Example

Statement:

Every prime number is odd.

False.

Counterexample:

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Since 2 is prime and even, the statement is false.

Example

Statement:

Every student scored above 90%.

A single student scoring below 90% disproves the statement.

3.12 Existential Quantifier

Symbol

$\exists$

Read as:

"There exists"

"There is at least one"

Form

$\exists x P(x)$

Meaning:

At least one element satisfies $P(x)$.

Example 1

There exists an integer x such that

$$x^2 = 25$$

True.

Because:

$$x = 5$$

and

$$x = -5$$

Example 2

There exists a prime number that is even.

True.

The number 2 satisfies the condition.

Example 3

There exists a natural number less than zero.

False.

No natural number is negative.

3.13 Universal vs Existential Quantifiers

Consider:

Every student passed.

This means:

All students satisfy the condition.

Some student passed.

This means:

At least one student satisfies the condition.

Important Difference

Universal statements require every element.

Existential statements require only one element.

3.14 Translating English Statements into Predicate Logic

Translation is one of the most important examination topics.

Example 1

Every student in this class has taken Java.

Let

$J(x)$: x has taken Java.

Translation:

For every student x ,

$J(x)$.

Example 2

Some student in this class has taken Java.

Translation:

There exists a student x such that

$J(x)$.

Example 3

Every prime number greater than 2 is odd.

Let

$P(x)$: x is prime

$O(x)$: x is odd

Translation:

For every x ,

if x is prime and $x > 2$,

then x is odd.

Example 4

There exists an even prime number.

Translation:

There exists x such that

x is prime and x is even.

3.15 Negation of Quantified Statements

Negating quantified statements is extremely important in proofs.

Negation of Universal Quantifier

Not (For every x , $P(x)$)

is equivalent to

There exists x such that not $P(x)$.

Example

Statement:

All students passed.

Negation:

At least one student failed.

Example

Statement:

Every integer is positive.

Negation:

There exists an integer that is not positive.

3.16 Negation of Existential Quantifier

Not (There exists x such that $P(x)$)

is equivalent to

For every x , not $P(x)$.

Example

Statement:

There exists a student who scored 100%.

Negation:

No student scored 100%.

Example

Statement:

There exists an even prime greater than 2.

Negation:

Every prime greater than 2 is not even.

3.17 Nested Quantifiers

A statement containing more than one quantifier is called a nested quantified statement.

Example 1

For every real number x ,

there exists a real number y such that

$$x + y = 0.$$

True.

Choose:

$$y = -x$$

Example 2

There exists a real number y such that

for every real number x ,

$$x + y = 0.$$

False.

No single value of y works for all x .

3.18 Order of Quantifiers

The order of quantifiers is extremely important.

In general,

"For every x there exists y "

is NOT equivalent to

"There exists y for every x "

Example

Every student has a favorite teacher.

For every student x ,

there exists a teacher y .

This may be true.

There exists a teacher who is everyone's favorite.

This is a completely different statement.

It may be false.

3.19 Applications of Predicate Logic

Predicate logic is used in:

1. Mathematics
 2. Computer Programming
 3. Artificial Intelligence
 4. Database Systems
 5. Software Verification
 6. Graph Theory
 7. Automata Theory
 8. Formal Language Theory
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Summary

In this chapter we studied:

- Predicate Logic
- Predicates and Propositional Functions
- Domain of Discourse
- Open Sentences
- Well-Formed Formulas
- Universal Quantifier
- Existential Quantifier
- Counterexamples
- Translation of English Statements
- Negation of Quantifiers
- Nested Quantifiers
- Order of Quantifiers

Predicate logic extends propositional logic and provides the foundation for expressing mathematical statements involving variables, relations, and quantifiers.